Key Features

Deterministic finality and settlement

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Arc's deterministic finality provides instant, irreversible transaction settlement in under one second.

Arc gives you deterministic, sub-second finality for every transaction. Once a block is committed, any transaction in that block is instantly and irreversibly final. This removes the uncertainty you may encounter on blockchains that rely on probabilistic finality.

Why deterministic finality matters

On proof-of-work or many proof-of-stake chains, transactions are considered final only after multiple confirmation blocks. Even then, there's a risk of chain reorganizations that can undo an arbitrary number of blocks and transactions therein.

With Arc, a transaction is either unconfirmed or final. There is no intermediate state. Once a transaction is included in a committed block, it cannot be reversed.

This allows you to build applications that demand high assurance, especially where financial risk must be minimized and operational standards are strict.

Sub-second confirmation

Arc's consensus engine, **Malachite**, finalizes blocks in less than one second. For comparison, Ethereum requires 12–15 minutes for finality, and many Layer-2 networks inherit similar delays from their settlement layer.

This speed enables use cases that are impractical on slower networks:

Use case	How finality helps
Point-of-sale payments	A merchant can confirm payment and release goods without waiting for additional block confirmations.
Cross-border settlement	Transfers between counterparties finalize instantly, eliminating the settlement windows that introduce counterparty risk.
Institutional clearing	Trades and margin calls settle with immediate certainty, matching the expectations of traditional financial infrastructure.
Composable workflows	Multi-step onchain flows (such as swap-then-bridge) can execute sequentially without polling or confirmation delays between steps.

Developer benefits

Deterministic finality simplifies application design by removing the edge cases that probabilistic chains force you to handle.

No reorg handling

You don't need retry logic, rollback mechanisms, or confirmation-count thresholds. A confirmed transaction stays confirmed.

Immediate offchain effects

Safely trigger downstream actions (webhooks, database writes, notifications) as soon as a block is committed, without waiting for additional confirmations.

Simplified state management

Your application only needs to track two transaction states — pending and final — rather than tracking a sliding confirmation window.

Enterprise compliance

Settlement finality is auditable and provable, meeting the assurance requirements of regulated financial institutions.

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